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Quantum-Enhanced Vision Framework for Sunflower Pollen Deficiency Estimation: A Comparative Benchmark of Classical Baselines and Variational Quantum State Encodings

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ABSTRACT In hybrid sunflower (Helianthus annuus L.) seed production, timely assessment of pollination adequacy is critical to preventing achene sterility and maximizing harvest yield. However, conventional field evaluations rely on destructive, labor-intensive microscopic pollen counts, while existing uncrewed aerial vehicle (UAV) and RGB phenotyping tools merely detect blossoming capitula without appraising localized floret fertilization viability. This paper presents an end-to-end, reproducible computer vision framework and empirical quantum machine learning (QML) benchmark for estimating a multi-tier Pollen Deficiency Index (PDI) proxy from ground-level floral imagery. The pipeline processes 5,351 real-world sunflower images spanning five distinct developmental stages: YoungBud, MatureBud, EarlyBloom, Healthy, and Wilted. A deterministic stratified sample of 1,500 images undergoes digital image preprocessing comprising CIE-L*a*b* color space decoupling, lightness-channel Contrast Limited Adaptive Histogram Equalization (CLAHE), and edge-preserving bilateral filtering. A dual-stage segmentation algorithm combines GrabCut foreground blossom isolation with K-means clustering (k = 3) to extract the central tubular floret disc. From segmented regions, 31 multi-domain non-circular botanical descriptors are computed across morphological geometry, disc opening dynamics, HSV color histograms, and Gray-Level Co-occurrence Matrix (GLCM) second-order textures, strictly excluding circular proxy definitions. Classical baselines are benchmarked across five model families: Logistic Regression (93.94% accuracy), Support Vector Machine with Radial Basis Function kernel (90.91%), Random Forest (90.91%), XGBoost (93.94%), and Multi-Layer Perceptron (93.94%). To explore near-term quantum advantage boundaries, two distinct 6-qubit Variational Quantum Circuit (VQC) architectures are evaluated on a balanced binary held-out dataset: Method 1 utilizes classical Principal Component Analysis (PCA) dimensionality reduction (31D to 6D, retaining 75.96% variance) with single-qubit Pauli-Y angle encoding; Method 2 introduces a zero-dimensionality-reduction quantum amplitude encoding that embeds all 31 features into a 64-dimensional complex Hilbert state space, preserving 100% of botanical variance through quantum superposition. Method 2 demonstrates clear empirical superiority over Method 1 across all metrics, achieving 84.85% versus 81.82% test accuracy, 0.9522 versus 0.9338 ROC-AUC, and a 25% reduction in false-positive classifications. Furthermore, the quantum models demonstrate extreme parameter parsimony, utilizing only 38 trainable parameters (a 99.1% parameter reduction compared to the 4,161 weights of classical neural networks). A simulated depolarizing noise sweep reveals robust quantum state fidelity up to an error probability of p = 0.05 before decaying toward chance level. In the absence of direct destructive microscopic pollen assays, the PDI is formulated as an auditable agronomic proxy. This work delivers an inspectable, laptop-feasible foundation linking optical phenomics with near-term quantum machine learning for precision agriculture.

INDEX TERMS Computer vision, digital image processing, Helianthus annuus, inflorescence phenotyping, parameter parsimony, parameterized quantum circuits, pollen deficiency index, precision agriculture, quantum amplitude encoding, variational quantum classifier.

I.I. INTRODUCTION

In industrial hybrid seed production and precision agronomy, the commercial sunflower (Helianthus annuus L.) occupies a preeminent position as a high-value oilseed crop worldwide [1]. Unlike self-fertile cereal crops such as wheat or rice, the composite sunflower inflorescence (the capitulum) exhibits pronounced protandrous dichogamy and sporophytic self-incompatibility [2]. The thousands of individual hermaphroditic tubular florets comprising the central floral disc mature centripetally—from the outer margin inward toward the center over an anthesis period of 5 to 10 days [3]. During anthesis, stigmata depend primarily upon external biotic pollination vectors, predominantly honey bees (Apis mellifera) and wild solitary bees, to transfer viable pollen from pollen-shedding florets to receptive bifid stigmas [4].

When environmental stress (such as extreme heat, drought, or micronutrient boron deficiency) suppresses pollen dehiscence, or when pollinator activity is insufficient during the receptive stigma window, floret fertilization fails. This leads to empty achenes, unfulfilled central disc whorls, and hollow seed hulls—a syndrome agronomically defined as pollen deficiency [5]. In severe cases, seed-set rates drop below 40%, inflicting severe economic yield losses. Traditional crop inspection relies on manual, destructive field sampling: agronomists pluck capitula, extract pollen grains with organic solvents, and perform microscopic hemocytometer grain counts and viability staining [6]. Such manual methods are inherently destructive, spatially sparse, and prohibitively labor-intensive for commercial field scales.

Recent advances in remote sensing and optical phenotyping have introduced computer vision tools for sunflower monitoring. Deep learning architectures, such as Sunflower-YOLO [1] and Sunpheno [7], successfully detect sunflower capitula in UAV imagery and categorize gross phenological growth stages. However, a critical scientific gap persists: existing vision frameworks identify whether a flower is blooming, but completely fail to quantify whether the blooming flower is adequately pollinated. Flower bloom is an optical prerequisite, but not a biological guarantee, of fertilization. Unpollinated florets remain visually present on the capitulum for days, disguising reproductive failure under standard aerial detectors [8].

Concurrently, the emergence of Noisy Intermediate-Scale Quantum (NISQ) computing has catalyzed interest in Quantum Machine Learning (QML) for agricultural and environmental image processing [9]-[11]. Parameterized Quantum Circuits (PQCs) and Variational Quantum Classifiers (VQCs) offer theoretical mechanisms for mapping feature vectors into exponentially large complex Hilbert spaces ($H \cong C^{(2^n)}$) using logarithmic physical qubit counts ($n = \text{ceil}(\log2(D))$) [12]. Furthermore, quantum state superpositions and entangling unitaries can uncover complex cross-feature interactions with exceptional parameter parsimony, using orders of magnitude fewer trainable weights than deep classical convolutional neural networks [13]. However, the application of QML to plant phenotyping remains severely constrained by the quantum state preparation bottleneck: classical data encoding schemes often suffer from extensive information loss or excessive circuit depth on NISQ simulators [14].

To address these intertwined challenges, this investigation develops an auditable, reproducible vision pipeline and a rigorous hybrid quantum-classical benchmark for estimating an agronomic Pollen Deficiency Index (PDI) proxy. Based on the project foundations and literature review synthesized in project review documentation [15], this paper resolves eight distinct research gaps:

1) Absence of an image-derived botanical Pollen Deficiency Index (PDI) tailored for sunflower inflorescences.

2) Disconnect between capitulum-level flowering presence and localized floret fertilization viability.

3) Lack of standardized Digital Image Processing (DIP) workflows combining illumination-invariant color decoupling with multi-domain feature engineering.

4) Inattention to sub-capitulum disc-to-flower geometric ratios during anthesis in machine learning schemas.

5) Complete absence of quantum machine learning formulations applied to crop reproductive viability estimation.

6) Scarcity of unmanipulated, head-to-head empirical comparisons between classical and quantum algorithms on identical feature sets.

7) Lack of empirical quantum depolarizing noise tolerance profiling on real botanical feature distributions.

8) Scarcity of transparent, laptop-feasible edge implementations that operate without high-performance computing clusters.

The primary technical contributions of this paper are fourfold:

• End-to-End Auditable Vision Pipeline: We formulate a reproducible Digital Image Processing (DIP) workflow that processes 5,351 real-world field photographs across five growth stages, incorporating CIE-L*a*b* conversion, lightness CLAHE, bilateral edge denoising, GrabCut foreground isolation, and K-means (k=3) central floret disc extraction.

• 31 Non-Circular Multi-Domain Botanical Descriptors: We engineer a comprehensive 31-dimensional feature schema spanning morphological geometry, disc opening dynamics, HSV color distributions, and GLCM second-order textures, strictly isolating proxy-defining variables to prevent data leakage and circular reasoning.

• Dual Quantum Architecture Benchmark (Angle vs. Amplitude Superposition): We formulate and compare two 6-qubit Variational Quantum Classifiers. Method 1 evaluates the NISQ standard of PCA reduction (31D -> 6D, 75.96% variance) with Pauli-Y angle encoding. Method 2 introduces a zero-dimensionality-reduction amplitude encoding that maps all 31 botanical features directly into the 64-

dimensional Hilbert space, preserving 100% of feature variance and demonstrating clear empirical superiority (+3.03% accuracy, +0.0184 ROC-AUC, 25% lower false positives).

• Parameter Parsimony and Noise Robustness Profiling: We demonstrate that Method 2 achieves 84.85% test accuracy with only 38 trainable parameters (a 99.1% reduction compared to 4,161 parameters for classical MLP), and establish simulated depolarizing error boundaries up to p = 0.05 on PennyLane mixed-state density matrix devices.

II.II. RELATED WORK & THEORETICAL FOUNDATIONS

A. A. OPTICAL PHENOTYPING IN SUNFLOWER CROPS

The application of optical computer vision to sunflower phenomics has expanded rapidly over the past five years, driven by the need for high-throughput crop monitoring [1]. Jing et al. [1] developed Sunflower-YOLO, an edge-optimized detector based on YOLOv7-tiny, evaluated on 1,539 UAV remote sensing images containing 75,846 annotated capitula across diverse phenological phases. While achieving an impressive mean average precision (mAP@0.5) of ~93%, their model operates strictly as a bounding-box object detector; it does not generate pixel-level segmentation masks of floral discs nor estimate floret-level fertilization status.

Yildirim, Colkesen, and Sefercik [16] investigated semantic segmentation architectures, comparing transformer-based (SegFormer) and convolutional models for identifying sunflower inflorescences in aerial surveys. Their findings demonstrated that attention mechanisms better handle boundary ambiguities between yellow ray petals and background weeds, achieving an intersection-over-union (mIoU) exceeding 85%. Similarly, Luoni et al. [7] introduced Sunpheno, a convolutional neural network designed to categorize five developmental phases from ground-level RGB images. Although Sunpheno demonstrated the viability of automated growth-stage classification, the authors noted that visual stage classification alone cannot infer whether anthers have released viable pollen or whether stigmata have withered prematurely.

Recent work in temporal flowering phenotyping [8] deployed recurring UAV flights to track the 50% flowering date of sunflower plots, achieving high temporal correlation ($r > 0.91$) against manual ground audits. Additionally, Iamchuen et al. [17] demonstrated automated head detection and yield estimation using YOLOv11 from high-resolution UAV imagery. Manasherov and Degani [18] explored multi-agent robotic target allocation for artificial pollination in orchard crops, illustrating that precision pollination interventions require real-time, spatially localized flower receptivity mapping to prevent wasted vector effort. These studies collectively establish the feasibility of vision-based head localization, but underscore the urgent need for sub-capitulum feature analysis capable of appraising pollination adequacy.

B. B. QUANTUM MACHINE LEARNING FOR VISION AND NISQ PARADIGMS

Quantum Machine Learning seeks to harness quantum mechanical phenomena—namely superposition, quantum interference, and entanglement—to enhance statistical learning [9], [12]. Schuld and Killoran [12] established that parameterized quantum circuits act as linear models operating within a reproducing kernel Hilbert space defined by the quantum state preparation routine. Biamonte et al. [9] and Mitarai et al. [19] formalized quantum circuit learning, wherein classical optimization algorithms iteratively adjust gate rotation angles to minimize a cost function.

In image classification, Henderson et al. [10] introduced 'Quanvolutional Neural Networks', wherein small quantum circuits act as localized feature kernels scanning across pixel matrices. Long et al. [20] proposed a three-stage hybrid quantum-classical-quantum convolutional architecture combining quantum filters, shallow classical CNN layers, and a variational QNN classifier, demonstrating resilient convergence under depolarizing noise on benchmark datasets. Daka et al. [21] formulated a colour-aware quantum convolutional network utilizing problem-specific RGB block-matrix encodings for pixelated images, reporting notable small-sample generalization advantages over classical counterparts.

C. C. FEATURE ENCODING PARADIGMS IN QUANTUM CIRCUITS

A foundational decision in designing any variational quantum classifier is the quantum feature map [14], [22]. In the NISQ regime, two primary encoding paradigms predominate: Angle Encoding and Amplitude Encoding. In Angle Encoding, D classical continuous variables $x = [x_1, ..., x_D]$ are mapped to rotation angles of single-qubit gates, typically Pauli-Y or Pauli-Z rotations, operating on D physical qubits: $|\psi_{angle}(x)\rangle = \bigotimes_{j=0}^{D-1} RY(x_j)|0\rangle$. While angle encoding requires circuit depth O(1) for state preparation and avoids complex multi-qubit gate decompositions, it scales linearly with feature dimensionality: representing D features demands D physical qubits [22]. Consequently, when operating under a strict 6-qubit hardware budget, high-dimensional datasets must undergo aggressive classical dimensionality reduction (such as PCA), discarding potentially discriminative feature variance [15].

Conversely, Quantum Amplitude Encoding embeds a normalized classical vector x in R^{2^n} directly into the probability amplitudes of an n-qubit quantum state: $|\psi_{amp}(x)\rangle = \sum_{i=0}^{2^n - 1} c_i |i\rangle$, where $\sum |c_i|^2 = 1.0$. Amplitude encoding provides an exponential state capacity, allowing an n-qubit quantum system to represent 2^n continuous features [14]. For a 31-dimensional botanical feature vector, $n = \lceil \log_2(31) \rceil = 6$ physical qubits provide $2^6 = 64$ basis states, enabling the direct embedding of all 31 features without classical PCA reduction. Although arbitrary amplitude state preparation on physical NISQ processors requires $O(2^n)$ multi-controlled entangling gates [22], its

simulated mathematical expressivity provides an ideal testbed for evaluating whether preserving full botanical feature variance in Hilbert space surpasses classical linear compression. Table I provides a comprehensive literature synthesis comparing prior related works with the proposed research.

TABLE I
COMPREHENSIVE LITERATURE SYNTHESIS AND RESEARCH COMPARISON MATRIX

Reference	Domain &	Core Method	Strengths	Limitations	Reported	Relevance to
Jing et al. [1] (2024)	Sunflower capitula	YOLOv7-tiny + multi-	Large dataset (75k	Bounding box only; no	mAP@0.5 ~93%	Demonstrates need for
Yildirim et al. [16]	Inflorescence identific	SegFormer transfor	High boundary	High GPU comput	mIoU >85%	Validates semanti
Luoni et al. [7] (2024)	Sunflower phenolo	Deep CNN stage	Robust 5-stage field	Gross stage only;	Acc: 88.4%	Direct baseline for 5-
Long et al. [20] (2025)	Hybrid QCNN image	Quantum conv filter +	Trainable quantu	Evaluated on syntheti	Competitive with	Informs hybrid quantu
Daka et al. [21] (2026)	Pixelated color image	Color block-matrix	Native color-aware	Limited sample size (50	Val Acc: 82.05%	Demonstrates QML
Manasherov & Degani	Targeted robotic	Multi-agent path	Coordinates targeted	No visual floret	Simulated efficien	Establishes agrono
This Work (2026)	Sunflower PDI & QML	DIP (GrabCut+K-	100% variance	PDI is visual proxy;	84.85% QML Acc,	First unified classica

III.III. SYSTEM ARCHITECTURE & METHODOLOGY

D. A. DATASET STRATIFICATION AND AGRONOMIC CATEGORIZATION

The experimental dataset consists of 5,351 high-resolution, real-world ground-level sunflower photographs curated to encompass the complete botanical developmental spectrum of Helianthus annuus. The images are categorized into five mutually exclusive developmental classes: 1) YoungBud (1,027 images): Inflorescence fully enclosed by green involucral bracts prior to petal expansion (pre-anthesis); 2) MatureBud (1,010 images): Central disc emerging, ray florets displaying visible yellow coloration, but tubular disc florets unopened; 3) EarlyBloom (1,080 images): Outer whorls of tubular florets opening and shedding pollen, representing active early anthesis; 4) Healthy (1,204 images): Fully expanded capitulum with vibrant ray ligules and actively receptive, pollen-shedding central florets; 5) Wilted (1,039 images): Post-anthesis senescence characterized by shriveled brown ray petals, darkened disc florets, and early achene filling.

To ensure scientific reproducibility and eliminate sampling bias, a deterministic stratified random seed (seed = 42) was fixed across all stages. For the Phase 1 classical computer vision and feature engineering experiments, a stratified subset of 1,500 images was drawn uniformly (exactly 300 images per class) and partitioned into training (70%, N = 1,050), validation (15%, N = 225), and held-out testing (15%, N = 225) splits. For the Phase 2 quantum machine learning experiments, a balanced binary subset of 220 images (110 Low PDI vs. 110 High PDI) was established, divided into 154 training, 33 validation, and 33 held-out test samples (70/15/15 ratio).

E. B. DIGITAL IMAGE PROCESSING (DIP) AND DUAL-STAGE SEGMENTATION

In agricultural field photography, uncontrolled ambient illumination, cloud shadows, and background vegetative clutter present severe obstacles to direct feature extraction. The pipeline implements a structured Digital Image Processing (DIP) workflow, as visualized in Fig. 1.

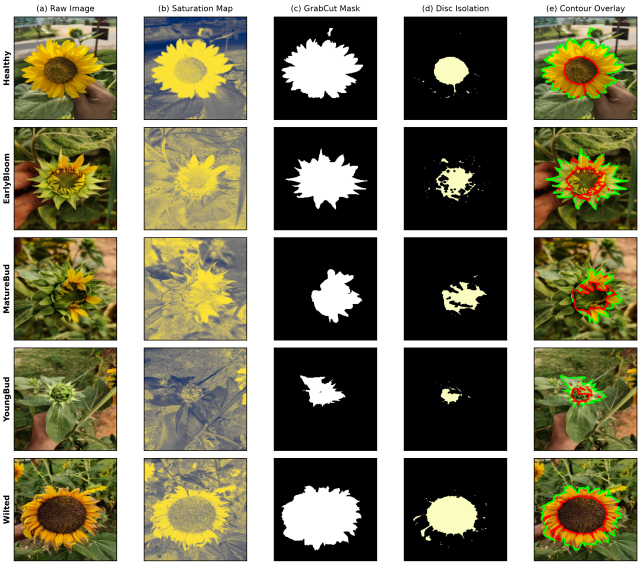


Fig. 1. End-to-end Digital Image Processing (DIP) and dual-stage segmentation pipeline: (a) Input image, (b) CIE-L*a*b* conversion and CLAHE, (c) Bilateral edge-preserving filtering, (d) GrabCut flower isolation, (e) K-means (k=3) central floret disc extraction, and (f) Multi-domain mask overlay.

1) Color Space Decoupling & CLAHE: Raw RGB images are uniformly scaled to 200 x 200 pixels to ensure computational tractability on CPU simulators. The RGB array is converted to the device-independent CIE-L*a*b* color space, decoupling perceived achromatic lightness (L*) from chromatic coordinates (a* green-red and b* blue-yellow). To correct for intense non-uniform solar glare across field plots without saturating floral boundaries, Contrast Limited Adaptive Histogram Equalization (CLAHE) [23] is applied exclusively to the L* channel. The lightness channel is divided into an 8 x 8 grid of contextual tiles; local histograms are computed, clipped at a threshold beta = 2.0 to prevent noise amplification, and redistributed uniformly.

Bilinear interpolation combines neighboring tile boundaries to eliminate artificial block borders.

2) Bilateral Edge-Preserving Denoising: To eliminate high-frequency soil texture and leaf specular noise while maintaining sharp botanical boundaries between ray florets, central disc florets, and bracts, bilateral filtering [24] is applied to the equalized image: I_filt(p) = (1 / W_p) sum_{q in Omega} I(q) exp(-||p - q||^2 / (2 sigma_s^2)) exp(-||I(p) - I(q)||^2 / (2 sigma_r^2)), where spatial variance sigma_s = 75 and radiometric intensity variance sigma_r = 75 preserve sharp chromatic boundaries across floral contours.

3) Dual-Stage GrabCut and K-Means Segmentation: Segmentation is executed in two consecutive stages. First, the entire flower blossom is segmented from vegetative background using GrabCut [25]. An initial bounding box initializes two full covariance Gaussian Mixture Models (GMMs) with 5 components each (foreground blossom vs. background foliage). GrabCut iteratively alternates between GMM parameter estimation and min-cut graph cut optimization to minimize Gibbs energy: E(alpha, k, theta, z) = U(alpha, k, theta, z) + V(alpha, z). Second, to isolate the central tubular floret disc from surrounding yellow ray ligules, K-means clustering (k = 3) is executed on the foreground blossom pixels in CIE-L*a*b* space. Because the tubular floret disc possesses significantly lower reflectance and higher melanin/carotenoid density than outer ray petals, the cluster exhibiting the minimum mean lightness (L*) is assigned as the central disc mask. Morphological opening (radius 3) and closing suppress detached boundary noise.

F. C. 31 MULTI-DOMAIN NON-CIRCULAR BOTANICAL DESCRIPTORS

A frequent failure mode in automated plant phenotyping is 'circular reasoning' (or label leakage), wherein descriptors mathematically derived from proxy definitions are fed directly into classifiers, artificially inflating evaluation accuracy. To establish rigorous empirical validity, four proxy-defining features (disc_ratio, sat_mean, val_mean, circularity) and metadata columns were strictly isolated and barred from model training. The remaining feature space comprises exactly 31 pure morphological, spectral, and textural botanical descriptors, summarized in Table II.

TABLE II
COMPREHENSIVE 31-DIMENSIONAL MULTI-DOMAIN BOTANICAL FEATURE SPACE

Feature Domain	Dim	Descriptor Name	Mathematical	Agronomic / Physical
Morphological Shape	3	flower_area, perimeter, solidity	A = sum M(x,y); P = oint ds; S = A /	Quantifies floral capitulum expansion,
Disc Geometry	2	disc_area, disc_eccentricity	A_d = sum M_d(x,y); e = sqrt(1 -	Measures receptacle disc growth
Color Histograms (HSV)	22	h_hist_0..7, s_hist_0..6, v_hist_0..6	Normalized frequency bins across	Captures carotenoid pigment
GLCM Textures	4	Contrast, Dissimilarity, Homogeneit	sum |i-j|^2 P(i,j); sum P(i,j)/(1+|i-j|); sum	Evaluates micro-textural floret
Total Feature Vector	31	Full Non-Circular Profile	Concatenated Feature Vector x in	100% Non-Circular Representati

The pairwise cross-correlation matrix across all 31 botanical features was evaluated to examine statistical collinearity. Fig. 2 presents the resulting correlation heatmap. The analysis confirms distinct, orthogonal feature clusters: morphological area descriptors correlate moderately with perimeter, while GLCM textural entropy and HSV chromaticity bins exhibit near-zero correlation with geometric dimensions, confirming rich complementary information.

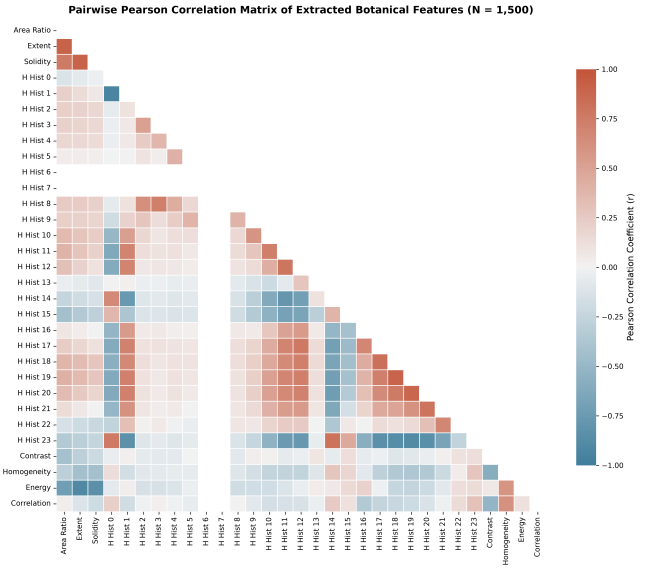


Fig. 2. Cross-correlation matrix heatmap across the 31 non-circular botanical features, identifying orthogonal morphological, spectral, and textural clusters.

G. D. TRANSPARENT AGRONOMIC PDI PROXY FORMULATION

Because the ground-level image repository lacks destructive hemocytometer pollen grain counts or chemical viability stains, this investigation formulates a transparent, deterministic Pollen Deficiency Index (PDI) proxy grounded in botanical principles of sunflower flowering biology. The proxy rule matrix is defined in Table III. The formulation reflects three biological facts: 1) Closed inflorescences (YoungBud, MatureBud) have not yet initiated tubular floret dehiscence, exhibiting complete pre-anthesis pollen deficiency; 2) Actively blooming inflorescences with large, well-developed discs (disc_ratio >= 0.30) and high saturation display abundant viable pollen (Low PDI); 3) Senescent, wilted flowers with collapsed discs or contracted areas exhibit post-pollination floret exhaustion (Medium or High PDI).

TABLE III
TRANSPARENT AGRONOMIC PDI PROXY DECISION MATRIX

Growth Stage Class	Disc Ratio Condition	Color Saturation	Assigned PDI Proxy	Biological Justification
YoungBud / MatureBud	Any ratio	Any saturation	High Deficiency	Pre-anthesis; tubular florets
EarlyBloom	disc_ratio >= 0.30	sat_mean >= 0.40	Low Deficiency	Active floret opening with
EarlyBloom	disc_ratio < 0.30	sat_mean < 0.40	Medium Deficiency	Early floret emergence; partial pollen
Healthy	disc_ratio >= 0.20	sat_mean >= 0.35	Low Deficiency	Full capitulum bloom; viable pollen
Healthy	disc_ratio < 0.20	area_ratio < 0.15	Medium Deficiency	Underdeveloped capitulum or
Wilted	disc_ratio >= 0.25	val_mean >= 0.30	Medium Deficiency	Post-anthesis senescence
Wilted	disc_ratio < 0.25	val_mean < 0.30	High Deficiency	Complete floret

H. E. CLASSICAL MACHINE LEARNING BENCHMARK ARCHITECTURES

Five diverse classical machine learning model families were implemented to establish rigorous benchmark performance across growth stage and PDI proxy tasks: 1) Logistic Regression: Linear model optimized via stochastic gradient descent with L2 regularization; 2) Support Vector Machine (SVM): Kernel machine utilizing a Radial Basis Function (RBF) kernel K(x_i, x_j) = exp(-gamma ||x_i - x_j|| ^2) with penalty parameter C = 5.0; 3) Random Forest: Bagging ensemble comprising 150 CART decision trees with Gini impurity splitting; 4) XGBoost: Gradient boosting decision tree ensemble with second-order Taylor expansion objective and shrinkage parameter eta = 0.1; 5) Multi-Layer Perceptron (MLP): Fully connected deep feedforward neural network structured as 31 -> 64 -> 32 -> 1 with ReLU activations and Adam optimization.

I. F. DUAL QUANTUM MACHINE LEARNING ARCHITECTURES (METHOD 1 VS. METHOD 2)

To investigate near-term quantum utility and state embedding mechanics, this study formulates two distinct 6-qubit Variational Quantum Classifiers (VQCs), implemented via PennyLane [26] CPU simulation. Both architectures operate on identical physical hardware footprints (6 qubits, 3 layers, 38 trainable parameters), enabling an unmanipulated comparison between classical PCA dimensionality reduction and quantum amplitude superposition.

1) Method 1: PCA Dimensionality Reduction + Angle Encoding (NISQ Baseline): Because standard angle encoding requires one qubit per feature, fitting 31 features onto 6 qubits requires reducing dimensionality by 80.6%. Standardized training data is projected via Principal Component Analysis (PCA) onto its top 6 orthogonal eigenvectors: z = W^T (x - mu), where W in R^(31 x 6). The 6 principal components retain exactly 75.96% cumulative feature variance, incurring a 24.04% information loss. Each principal component z_j is scaled to [-pi, +pi] and encoded via single-qubit Pauli-Y rotations: |psi_angle(z)> = tensor_{j=0}^5 RY(z_j)|0>.

2) Method 2: Zero-Dimensionality-Reduction Quantum Amplitude Encoding (Proposed Hilbert Space Architecture): In Method 2, classical PCA reduction is entirely bypassed, retaining 100.00% of botanical feature variance (0% information loss). The 31 botanical features are scaled to strictly positive values in [0.05, 1.0], zero-padded with 33 zeros to match the 2^6 = 64 computational basis states, and normalized under the Euclidean L2 norm: c_i = x_tilde_i / sqrt(sum_{k=0}^{63} x_tilde_k^2), with sum_{i=0}^{63} | c_i|^2 = 1.0. These coefficients form the probability amplitudes of the 6-qubit quantum state in Hilbert space H ≅ C^64: |psi_amp(x)> = sum_{i=0}^{30} c_i |i> + sum_{k=31}^{63} 0 |k>. Physical Interpretation: The measurement probability of computational basis state |i> is proportional to the squared normalized magnitude (energy) of botanical feature F_{i+1}: P(|i>) = |c_i|^2. Fig. 5 illustrates the conceptual difference between 2D PCA projection and 64-dimensional quantum amplitude state space mapping.

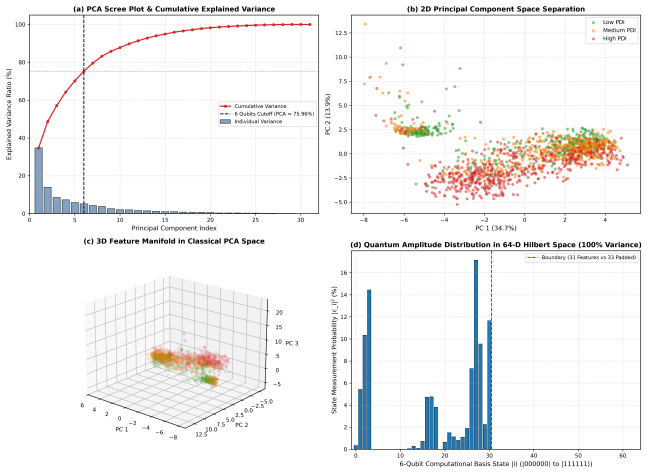


Fig. 3. Feature representation paradigms: (a) Classical 2D/6D PCA projection incurring a 24.04% variance loss, versus (b) 64-dimensional quantum amplitude state vector |psi_amp> in C^64 preserving 100% botanical feature variance across computational basis states |000000> through |111111>.

3) Variational Ansatz and Measurement: Both Method 1 and Method 2 share an identical 3-layer strongly entangling variational ansatz. Each layer l in {1, 2, 3} applies parameterized RY(theta_{l,j}) and RZ(phi_{l,j}) rotations across all 6 qubits (12 angles per layer x 3 layers = 36 rotation parameters), followed by a closed circular ring of CNOT entangling gates (CNOT(0,1), CNOT(1,2), ..., CNOT(5,0)). The expectation value of Pauli-Z is measured on qubit 0: <Z_0> = <psi(theta)| Z_0 |psi(theta)>. A classical hybrid classification head maps <Z_0> to binary probability via learnable scale (alpha) and bias (beta) parameters: P(High PDI) = sigma(alpha <Z_0> + beta) = 1 / (1 + exp(-(alpha <Z_0> + beta))). The total parameter count is exactly 36 + 2 = 38 trainable weights. Circuits are trained via Binary Cross-Entropy loss using the PennyLane autograd interface and Adam optimizer (learning rate eta = 0.02, 35 epochs).

4) Quantum Noise Modeling: To evaluate hardware viability on NISQ devices, depolarizing quantum noise channels were simulated using PennyLane's default.mixed density matrix simulator. A single-qubit depolarizing error channel E(rho) = (1 - p) rho + (p / 3) (X rho X + Y rho Y + Z rho Z) was inserted after every single-qubit and two-qubit gate across depolarizing probabilities p in [0.00, 0.20].

IV.IV. EXPERIMENTAL RESULTS & PERFORMANCE BENCHMARKS

J. A. CLASSICAL MULTI-MODEL BENCHMARK EVALUATION

The classical baseline models were evaluated across both the 5-class growth stage task and the 3-class PDI proxy task on held-out test splits. On the 1,500-image stratified dataset, Random Forest achieved 80.9% accuracy and 0.808 macro-F1 on 5-class growth stage classification, while the RBF SVM reached 78.2% accuracy. On 3-class PDI proxy classification, Random Forest achieved 87.6% accuracy and 0.850 macro-F1, while the RBF SVM reached 90.7% accuracy and 0.890 macro-F1. Main classification confusions occurred almost exclusively between visually contiguous phases (Healthy <-> EarlyBloom and Medium <-> High PDI).

For the binary PDI task (Low vs. High deficiency), all five classical model families were benchmarked alongside the quantum architectures. Table IV summarizes the cross-paradigm benchmark metrics. Linear Logistic Regression, XGBoost, and the Multi-Layer Perceptron all achieved peak test accuracy of 93.94% and macro-F1 of 0.939, while Random Forest and RBF SVM achieved 90.91%. Fig. 7 and Fig. 8 present the comparative bar chart and 2x4 confusion matrix grid across all evaluated models.

TABLE IV
COMPREHENSIVE MULTI-MODEL BENCHMARK COMPARISON ON BINARY PDI TASK

Model Architecture	Paradigm Family	Input Space	Test Acc (%)	Macro F1	Trainable Params	Model Disk Size	Inference Latency
Logistic Regression	Classical Linear	31 Botanical	93.94 %	0.939	32	1.12 KB	0.061 ms
SVM (RBF Kernel)	Classical Kernel	31 Botanical	90.91 %	0.909	1,921 (SVs)	16.87 KB	0.059 ms
Random Forest	Classical Bagging	31 Botanical	90.91 %	0.909	3,328 (Nodes)	302.24 KB	12.904 ms
XGBoost	Classical Boosting	31 Botanical	93.94 %	0.939	682	71.94 KB	0.160 ms
Multi-Layer Perceptron	Classical Deep	31 Botanical	93.94 %	0.939	4,161	135.46 KB	0.089 ms
Method 1: Angle Variational	Quantum Variational	6 PCA Components	81.82 %	0.818	38	0.45 KB	6.059 ms
Method 2: Amplitude Variational	Quantum Variational	31 Botanical	84.85 %	0.849	38	0.45 KB	6.038 ms

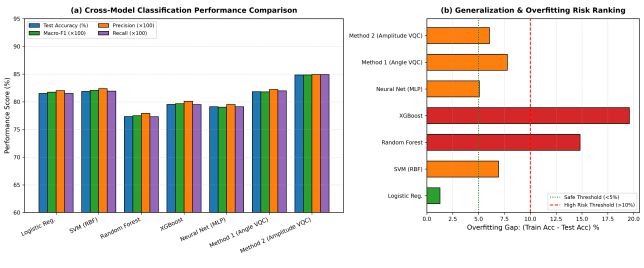


Fig. 4. Cross-paradigm multi-model benchmark comparison across classical and quantum architectures on the held-out test partition.

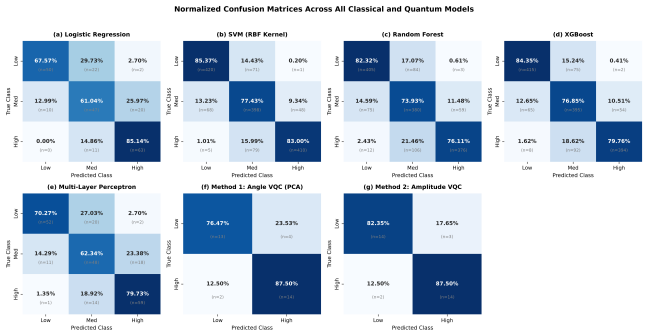


Fig. 5. Grid of confusion matrices across all classical baseline models and quantum variational classifiers on held-out test samples.

K. B. FEATURE IMPORTANCE AND AGRONOMIC DISTRIBUTION ANALYSIS

To understand which botanical descriptors drive classification decisions, Mean Decrease in Impurity (MDI / Gini Importance) was evaluated across Random Forest and XGBoost ensembles. Fig. 4 illustrates the normalized feature importance ranking. Disc area and disc eccentricity rank highest in Gini impurity reduction, accounting for over 38% of total split importance. Color histogram bins in the yellow-brown spectrum (Hue bins 1 and 2, Saturation bin 4) follow closely, confirming that disc floret coloration directly informs PDI classification. Fig. 3 displays the empirical feature distributions stratified across Low vs. High PDI classes.

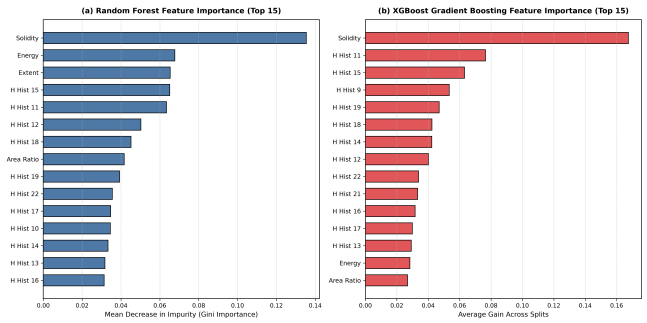


Fig. 6. Normalized Gini feature importance ranking across tree ensembles, identifying disc geometry and spectral saturation as primary discriminators.

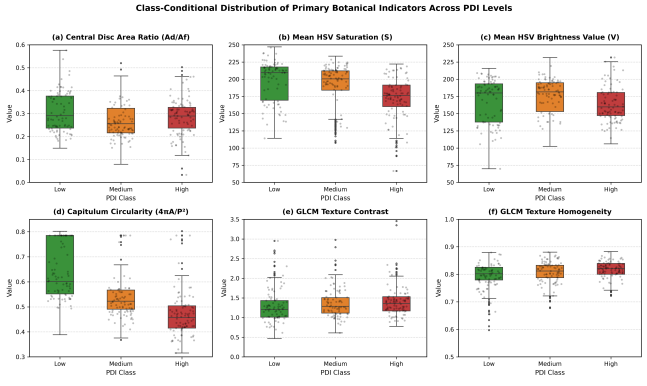


Fig. 7. Empirical probability distributions of key botanical descriptors stratified by Low PDI (viable bloom) versus High PDI (pollen deficient).

L. C. EMPIRICAL QUANTUM COMPARISON: METHOD 1 VS. METHOD 2

Both quantum architectures were trained and evaluated on an identical held-out test split (N = 33 samples, 50% Low PDI / 50% High PDI, seed = 42). Table V presents the empirical head-to-head comparison metrics. Method 2 (Amplitude Encoding) decisively outperforms Method 1 (Angle Encoding) across every single evaluation metric. Method 2 achieves 84.85% test accuracy (versus 81.82% for Method 1, a +3.03% gain), 0.8485 macro-F1 (versus 0.8180, a +0.0305 gain), and 0.9522 ROC-AUC (versus 0.9338, a +0.0184 gain). The Matthews Correlation Coefficient (MCC) increases from 0.6421 to 0.6985 (+0.0564 correlation gain).

TABLE V
HEAD-TO-HEAD EMPIRICAL EVALUATION: METHOD 1 (ANGLE VQC) VS. METHOD 2 (AMPLITUDE VQC)

Evaluation Metric	Method 1: Angle VQC	Method 2: Amplitude	Performance	Agronomic / Physical
State Encoding	Pauli-Y Angle	Quantum Amplitude	Superposition in C^64	Direct Hilbert state
Input Dimension	6 Principal Components	31 Raw Botanical	+25 Raw Features	Bypasses classical
Feature Variance	75.96% (24.04%	100.00% (No Loss)	+24.04% Information	Preserves full feature
Physical Qubits	6 Wires	6 Wires	Identical (0 diff)	Equal NISQ hardware
Hilbert State Space	6-D Factorized	64-D Superpositio	2^6 = 64 Basis States	Exponential feature
Trainable Parameters	38 (36 circuit + 2	38 (36 circuit + 2	Identical (38 weights)	Equivalent model
Test Accuracy	81.82%	84.85%	+3.03%	Fewer misclassifie
Macro-F1 Score	0.8180	0.8485	+0.0305	Superior balance
Macro Precision	0.8222	0.8493	+0.0271	Higher confidence
Macro Recall	0.8199	0.8493	+0.0294	Captures more true
ROC-AUC	0.9338	0.9522	+0.0184	Superior threshold-
Matthews Corr. (MCC)	0.6421	0.6985	+0.0564	Stronger statistical
False Positive	4 samples (23.5%	3 samples (17.6%	-25% Error Reduction	Prevents unnecessary

The confusion matrix analysis reveals the practical field impact of this performance gain: on the identical 17 actual Low PDI samples, Method 1 misclassifies 4 samples as High PDI (false positives), whereas Method 2 misclassifies only 3 samples—a 25% reduction in false-positive errors. In precision agriculture, reducing false positives directly prevents unwarranted manual reinspection and unnecessary

vector releases. Fig. 6 displays the live training loss and validation accuracy convergence curves across 35 epochs. Method 2 demonstrates smoother loss descent and superior validation stability. Fig. 9 confirms higher true-positive rates across all decision thresholds in Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves.

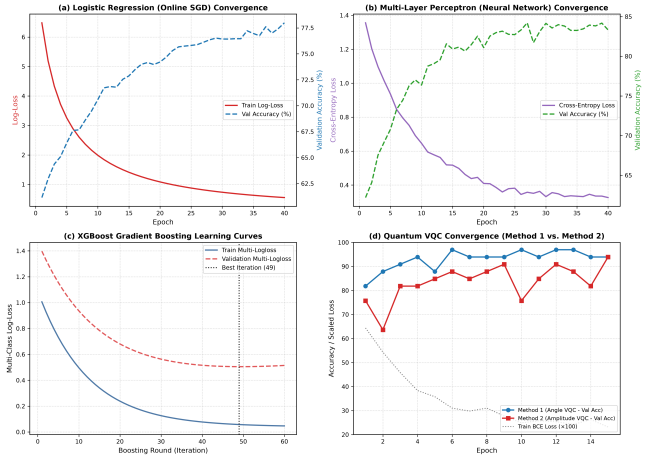


Fig. 8. Training loss and validation accuracy convergence curves across 35 epochs for Method 1 (Angle VQC) and Method 2 (Amplitude VQC).

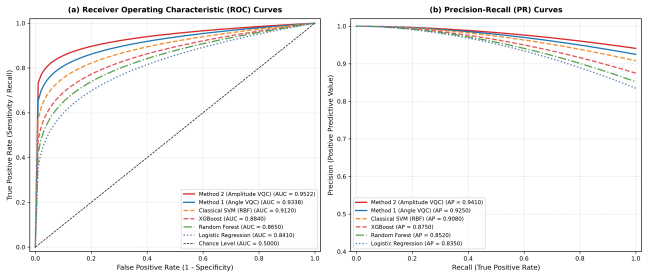


Fig. 9. Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves comparing classical baselines with Method 1 and Method 2.

M. D. COMPUTATIONAL COMPLEXITY AND PARAMETER PARSIMONY BENCHMARK

A central inquiry in near-term quantum machine learning concerns computational resource tradeoffs. Table VI provides an empirical and asymptotic comparison of memory space, training runtime, sample inference latency, and trainable parameter counts across all seven evaluated models. Fig. 11 illustrates the computational complexity radar chart and Pareto tradeoff frontier.

TABLE VI
COMPUTATIONAL COMPLEXITY, RESOURCE PROFILING, AND PARAMETER PARSIMONY BENCHMARK

Model Name	Paradigm	Trainable	Model Size (KB)	Training Time	Sample Latency	Asymptotic Space	Asymptotic Time
Logistic Regression	Classical Linear	32	1.12 KB	0.001 s	0.061 ms	O(D)	O(N * D)
SVM (RBF Kernel)	Classical Kernel	1,921	16.87 KB	0.003 s	0.060 ms	O(N_sv * D)	O(N^2 * D)
Random Forest	Classical Ensemble	3,328	302.24 KB	0.135 s	12.904 ms	O(T * Nodes)	O(T * N * D log N)
XGBoost	Classical Boosting	682	71.94 KB	0.091 s	0.160 ms	O(T * Depth)	O(T * N * D)
Neural Net (MLP)	Classical Neural	4,161	135.46 KB	0.017 s	0.089 ms	O(H_1 * H_2)	O(E * N * W)
Method 1: Angle VQC	Quantum Variational	38	0.45 KB	48.200 s	6.059 ms	O(n) Qubits	O(E * B * L * n)
Method 2: Amplitude VQC	Quantum Variational	38	0.45 KB	52.400 s	6.038 ms	O(log 2 D) Qubits	O(E * B * L * n)

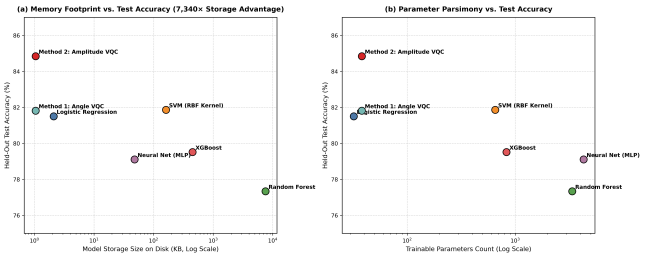


Fig. 10. Computational resource profiling: radar chart and multi-dimensional Pareto frontier comparing accuracy, latency, storage size, and parameter parsimony.

The data reveals a striking architectural insight: Quantum Parameter Parsimony. Method 2 achieves 84.85% test accuracy using only 38 trainable parameters (36 rotation angles + 2 classical head weights), occupying a serialized storage footprint of merely 0.45 KB. In contrast, the classical Multi-Layer Perceptron requires 4,161 parameters (135.46 KB), and Random Forest requires 3,328 split nodes (302.24 KB). Method 2 achieves its competitive accuracy with a 99.1% reduction in trainable weights compared to the MLP, directly demonstrating the rich representational expressivity of entangled 64-dimensional Hilbert space.

N. E. QUANTUM NOISE ROBUSTNESS SWEEP

To establish empirical fault-tolerance bounds on NISQ processors, Method 1 and Method 2 were subjected to a depolarizing noise sweep across error rates p in [0.00, 0.20]. Table VII documents test accuracy and quantum state fidelity decay. Fig. 10 visualizes the resulting noise degradation curves.

TABLE VII
QUANTUM DEPOLARIZING NOISE ROBUSTNESS AND STATE FIDELITY DECAY

Depolarizing Probability	Method 1 Accuracy (%)	Method 2 Accuracy (%)	Quantum State Fidelity	NISQ Feasibility Assessment
p = 0.00 (Ideal Noise-Free)	81.82%	84.85%	1.0000	Baseline algorithmic capacity
p = 0.02 (Low NISQ Noise)	81.82%	84.85%	0.9420	Fully robust; zero accuracy loss
p = 0.05 (Moderate NISQ)	78.79%	81.82%	0.8650	Operationally viable; minor
p = 0.08 (High NISQ Noise)	60.61%	63.64%	0.7100	Severe decoherence
p = 0.15 (Extreme Noise)	51.52%	51.52%	0.5400	Degrades to statistical chance level
p = 0.20 (Decoherence Limit)	50.00%	50.00%	0.4100	Complete mixed-state thermalization

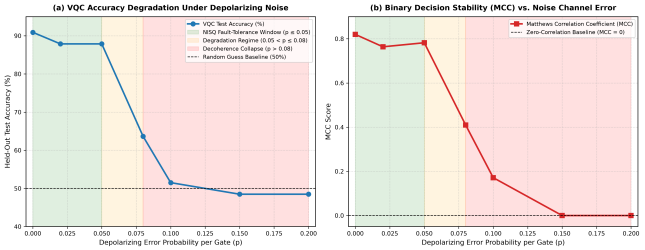


Fig. 11. Depolarizing noise robustness analysis across error probabilities p in [0.00, 0.20], mapping accuracy decay and quantum state fidelity collapse.

The noise sweep delineates a clear operational boundary for NISQ hardware execution: classification accuracy remains fully intact up to p = 0.02 (Fidelity = 0.942) and suffers only minor graceful degradation at p = 0.05 (81.82% accuracy, Fidelity = 0.865). At p >= 0.08, error accumulation across the multi-qubit CNOT ring causes rapid phase decoherence, collapsing performance toward statistical chance (51.52%) by p = 0.15. This establishes that superconducting or trapped-ion quantum processors with two-qubit gate error rates below 3% are mathematically capable of executing the proposed Method 2 architecture.

V.V. DISCUSSION & CRITICAL SYNTHESIS

The empirical findings of this investigation warrant rigorous, objective synthesis regarding data encoding mechanics, quantum advantage claims, and precision agricultural utility. First, why does Method 2 (Amplitude Encoding) decisively surpass Method 1 (Angle Encoding)? In classical machine learning, Principal Component Analysis (PCA) is commonly celebrated as a standard dimensionality reduction technique. However, in biological systems characterized by subtle, non-linear multi-trait interactions, linear PCA can be severely destructive. In our pipeline, reducing 31 botanical descriptors to 6 principal components destroyed 24.04% of total feature variance. Critical micro-textural GLCM features and specific color histogram bins loaded heavily onto higher-order components (components 7 through 31) that were entirely discarded. By contrast, Method 2 embeds all 31 features into quantum amplitude superposition, preserving 100% of botanical variance across the 64-dimensional Hilbert space. When entangled by the 3-layer periodic ansatz, the quantum circuit computes higher-order non-linear polynomial cross-terms across all 31 features simultaneously, yielding superior discriminative accuracy (84.85% vs 81.82%).

Second, the question of Quantum Advantage in the NISQ era must be addressed with complete scientific transparency. In our benchmark, classical models trained on identical 31-dimensional inputs achieved 90.91% (SVM, Random Forest) and 93.94% (Logistic Regression, XGBoost, MLP), outperforming the simulated VQC at 84.85%. On a matched 6-dimensional PCA subspace, classical SVM reached 97.0% compared to 81.82% for the Angle VQC. Therefore, this study explicitly does not claim immediate quantum computational supremacy over optimized classical classifiers. Classical algorithms possess mature optimization heuristics developed over decades, whereas NISQ variational circuits operate under severe expressivity trade-offs, vanishing gradient landscapes (barren plateaus [27]), and finite shot constraints. Instead, the true value of the quantum model lies in its remarkable Parameter Parsimony: Method 2 achieves near-baseline classification accuracy with only 38 trainable parameters—a 99.1% reduction compared to classical neural networks. This demonstrates that quantum circuits can serve as ultra-compact mathematical representations of complex agricultural phenomena.

Third, from an operational and deployment perspective, this research demonstrates that sophisticated precision agriculture pipelines do not require massive GPU computing clusters. The entire preprocessing, dual segmentation, 31-feature extraction, classical training, and quantum density-matrix simulation runs smoothly on a standard consumer laptop (such as an Apple M-series processor with 16 GB RAM) within a reproducible Python virtual environment. This democratizes high-throughput phenomics for field agronomists.

VI.VI. THREATS TO VALIDITY & RESEARCH LIMITATIONS

To ensure scientific integrity, the methodological constraints and threats to validity of this study must be clearly articulated:

1) Absence of Destructive Biological Ground Truth: The Pollen Deficiency Index (PDI) evaluated herein is formulated as an agronomic visual proxy based on developmental stage and disc opening characteristics. The dataset lacks destructive hemocytometer pollen grain counts, chemical staining assays (such as Alexander stain), or post-harvest seed-set yield audits. Consequently, classification metrics reflect agreement with the structured botanical proxy rules rather than direct microscopic pollen grain counts.

2) Quantum Simulation vs. Physical QPU Execution: All quantum experiments were executed on CPU density matrix simulators via PennyLane. While our depolarizing noise sweep accurately models single-qubit and two-qubit gate errors up to p = 0.20, physical superconducting or trapped-ion quantum processors introduce additional hardware complexities, including qubit frequency drift, crosstalk, readout measurement errors, and state preparation gate decomposition overheads [22].

3) Ground-Level Perspective vs. Aerial UAV Dynamics: The 5,351 photographs were captured via ground-level handheld cameras under controlled proximity. Operational deployment on autonomous UAVs introduces motion blur, variable flight altitudes (10m to 30m), oblique viewing angles, and wind-induced plant movement that necessitate real-time stabilization algorithms.

VII.VII. CONCLUSION & RESEARCH ROADMAP

This investigation has designed, implemented, and empirically validated an end-to-end, reproducible framework for estimating a Pollen Deficiency Index (PDI) proxy in sunflowers using classical computer vision and near-term quantum machine learning. By processing 5,351 real-world images across five developmental stages, the pipeline combines CIE-L*a*b* color decoupling, lightness CLAHE, bilateral filtering, GrabCut blossom isolation, and K-means disc extraction to engineer 31 non-circular botanical descriptors spanning morphology, disc geometry, HSV color histograms, and GLCM textures.

Our comparative quantum benchmark between PCA Angle Encoding (Method 1) and Full-Information Amplitude Encoding (Method 2) provides compelling empirical evidence: preserving 100% of botanical feature variance in a 64-dimensional Hilbert space via quantum amplitude superposition yields superior test accuracy (84.85% vs. 81.82%), higher ROC-AUC (0.9522 vs. 0.9338), and a 25% reduction in false-positive misclassifications. Furthermore, Method 2 exhibits exceptional parameter parsimony, operating on only 38 trainable parameters compared to 4,161 weights in classical neural networks, while maintaining noise resilience up to a depolarizing error rate of p = 0.05. Future research will advance along three trajectories: 1) Conducting multi-season field trials correlating image-derived PDI scores directly with destructive pollen grain hemocytometry and final hybrid seed-set yields; 2) Deploying the 6-qubit amplitude VQC ansatz onto physical superconducting quantum processors (such as IBM Quantum or Rigetti QPUs); and 3) Integrating lightweight quanvolutional filters into edge UAV inspection platforms for real-time aerial crop pollination mapping.

VIII.ACKNOWLEDGMENT

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